A Few Notes on Maxwell's Equations

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1 Introduction

Maxwell's equations are a set of four fundamental equations that describe the behavior of electric and magnetic fields in classical electromagnetism. They were formulated by the Scottish physicist James Clerk Maxwell in the 19th century and are crucial in understanding the behavior of electromagnetic waves, such as light (an idea previously articulated by Faraday [1]). Maxwell's equations are typically written in both differential and integral forms [16].

Maxwell's equations are fundamental to classical electromagnetism and have played a crucial role in the development of modern physics and technology, including the understanding of electromagnetic waves, the design of electrical circuits, and the development of technologies such as radio, television, and wireless communication, to name a few.

The rest of this note is organized as follows: Section 2 provides a few definitions needed to discuss Maxwell's equations. Section 3 discusses the differential form of Maxwell's equations, and Section 4 discusses the integral form of Maxwell's equations. Section 5 discusses a few of the properties of electromagnetic waves. Section 6 briefly discusses Oliver Heaviside's contributions to Maxwell's equations. Finally, Section 7 outlines some conclusions and final thoughts.

2 Definitions

Symbol	**What**	**Dimensions**	**Notes**
$\mathbf{B}$	Magnetic field	Newton/m/Ampere	$1\ \mathrm{N} = 1\ \mathrm{kg} \cdot \mathrm{m} \cdot \mathrm{s}^{-2}$
$\mathbf{E}$	Electric field	Newton/Coulomb	$1\ \mathrm{C} = 1\ \mathrm{A} \cdot \mathrm{s}$
$\mathbf{J}$	Current density	Ampere/m^2	$1\ \mathrm{A} = 6.241509074 \times 10^{18}$ e/s
ρ	Charge density	Coulomb/m^3	$1\ \mathrm{C} = 6.241509074 \times 10^{18}$ e
c	Speed of light in a vacuum	m/s	$c = 2.99792458 \times 10^8$ m/s
ϵ_0	Permittivity of free space	Farad/m	$1\ \mathrm{F} = 1\ \mathrm{kg}^{-1} \cdot \mathrm{m}^{-2} \cdot \mathrm{s}^4 \cdot \mathrm{A}^2$
μ_0	Permeability of free space	Henry/m	$\mu_0 = 4\pi \times 10^{-7}$ H/m

Table 1: A Few Definitions

2.1 Electric and Magnetic Constants in Maxwell's Equations

Maxwell's equations are usually defined using three constants: the speed of light (c), the electric permittivity of free space (ϵ_0), and the magnetic permeability of free space (μ_0).

To calculate the value of these constants, first note that the magnetic permeability of free space μ_0 is taken to have the exact value $\mu_0 = 4\pi \times 10^{-7}$ H/m (see Table 1). So μ_0 is given[1]. We also know that the speed of light c is equal to 2.99792458×10^8 m/s so we also can consider c as given.

Next, notice ϵ_0 and μ_0 are not independent but rather are related to the speed of light by the following equation [16]:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \tag{1}$$

Since we know the values of μ_0 and c, we can find the value of ϵ_0 by solving for it in Equation (1):

$$\epsilon_0 = \frac{1}{\mu_0 c^2} \tag{2}$$

Substituting the values of μ_0 and c into Equation (2) we get that $\epsilon_0 = 8.8541878128 \times 10^{-12}$ F/m.

Finally, note that in the presence of polarizable or magnetic media [18], the effective constants will have different values. In the case of a polarizable medium, called a dielectric, the comparison is stated as a relative permittivity or a dielectric constant. In the case of magnetic media, the relative permeability may also be stated.

Back to definitions. For some field $\mathbf{F}$, we call $\nabla \cdot \mathbf{F}$ the divergence of $\mathbf{F}$, while $\nabla \times \mathbf{F}$ is called the curl of $\mathbf{F}$ [3]. It is worth noting that the standard dot (or inner) product is defined as follows:

Definition 2.1. Standard Inner Product: $\mathbf{a} \cdot \mathbf{b}$

The inner product (aka dot product or scalar product) of two n-dimensional vectors $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$, usually denoted by either $\langle \mathbf{a}, \mathbf{b} \rangle$ or $\mathbf{a} \cdot \mathbf{b}$, is defined to be the scalar value

$$\mathbf{a} \cdot \mathbf{b} := \sum_{j=1}^{n} a_j b_j$$

Since a_j and b_j are scalars (and scalar multiplication commutes) the inner product commutes:

$$\mathbf{a} \cdot \mathbf{b} = \sum_{j=1}^{n} a_j b_j = \sum_{j=1}^{n} b_j a_j = \mathbf{b} \cdot \mathbf{a} \tag{3}$$

This is all good, but here we are using the dot notation for the divergence and the divergence is missing a key property that the standard dot product has, namely, commutivity (Equation (3)). Specifically, the standard dot product commutes but the divergence does not:

$$\nabla \cdot \mathbf{F} \neq \mathbf{F} \cdot \nabla \tag{4}$$

[1] 1 Henry/m = 1 Newton/Ampere^2 so $\mu_0 = 4\pi \times 10^{-7}$ H/m $\approx 1.25663706143 \times 10^{-6}$ N/A^2.

One way to see this is to notice that the LHS of Equation (4) is a scalar function while the RHS is a differential operator.

One other set of definitions we will need concerns the unit vectors in $\mathbb{R}^3$. Specifically, $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are defined to be the unit vectors in $\mathbb{R}^3$ in the x, y, and z directions respectively [2]. Note that $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are the canonical basis vectors for $\mathbb{R}^3$ and have column vector format

$$\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \ \hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \ \hat{\mathbf{k}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

3 Maxwell's Equations (differential form)

The differential form of Maxwell's equations assumes the absence of magnetic or polarizable media since these media can change the values of the permittivity and the permeability (see Section (2.1)). The differential form of Maxwell's equations is:

1. **Gauss' Law for Electricity**

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

   Gauss' law for electricity states that a stationary charge (ρ) creates an electric field ($\mathbf{E}$) [11].

2. **Gauss' Law for Magnetism**

$$\nabla \cdot \mathbf{B} = 0$$

   Gauss' law for electricity states that magnetic monopoles can not exist [7].

3. **Faraday's Law of Electromagnetic Induction**

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

   Faraday's law of electromagnetic induction states that a time-varying magnetic field ($\mathbf{B}$) creates an electric field ($\mathbf{E}$) [8].

4. **Ampere's Law with Maxwell's addition**

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

   Ampere's law with Maxwell's addition states that flowing electric current ($\mathbf{J}$) or a time-varying electric field ($\mathbf{E}$) creates a magnetic field ($\mathbf{B}$) [12].

4 Maxwell's Equations (integral form)

1. **Gauss' Law for Electricity**

$$\oint_C \mathbf{E}\, d\mathbf{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

where Q_{enclosed} is the enclosed charge. This equation implies that the electric field lines do not form a continuous closed path. Said another way: isolated positive or negative charges can exist.

2. **Gauss' Law for Magnetism**

$$\oint_C \mathbf{B}\, d\mathbf{A} = 0$$

This equation says that magnetic lines of force form a continuous closed path. In other words, it means that no isolated magnetic monopole can exist.

3. **Faraday's Law of Electromagnetic Induction**

$$\oint_C \mathbf{E}\, d\mathbf{l} = -\frac{d}{dt}\Phi_{\mathbf{B}}$$

where $\Phi_{\mathbf{B}}$ is the magnetic flux through a closed path. This equation says that the line integral of the electric field around any closed path is equal to the rate of change of magnetic flux through the closed path bounded by the surface.

4. **Ampere's Law with Maxwell's addition**

$$\oint_C \mathbf{B}\, d\mathbf{l} = \mu_0 I_{\text{enclosed}} + \mu_0\epsilon_0 \frac{d}{dt}\int_S \mathbf{E}\, d\mathbf{A}$$

where I_{enclosed} is the enclosed current. This equation says that both conduction and displacement current can produce a magnetic field.

5 Electromagnetic Waves

Classical electromagnetic waves, often referred to simply as electromagnetic waves, are a fundamental concept in classical physics that describes the behavior of electromagnetic radiation, including light, without considering the quantum mechanical nature of light (as described by quantum electrodynamics). These waves are described by classical electromagnetism, primarily through the equations developed by James Clerk Maxwell in the 19th century [16].

The key characteristics and features of classical electromagnetic waves include:

- **Wave Equation**: The behavior of classical electromagnetic waves is described by Maxwell's equations, which consist of four fundamental equations (see Sections 3 and 4):
  1. Gauss' Law for Electricity
  2. Gauss' Law for Magnetism

3. Faraday's Law of Electromagnetic Induction
4. Ampère's Law with Maxwell's addition

- **Wave Propagation**: Classical electromagnetic waves are solutions to Maxwell's equations and can propagate through space as self-sustaining waves of electric and magnetic fields. These waves travel at the speed of light in a vacuum, c, which is approximately 3.0×10^8 m/s.
- **Transverse Waves**: Electromagnetic waves are transverse waves, which means that the oscillations of the electric and magnetic fields are perpendicular to the direction of wave propagation. This is depicted in Figure 1. Contrast with longitudinal waves, where the oscillations are along the direction of propagation.
- **Continuous Spectrum**: Classical electromagnetic waves encompass a continuous spectrum of frequencies and wavelengths, from extremely low-frequency radio waves to extremely high-frequency gamma rays. Each portion of the spectrum corresponds to different types of electromagnetic radiation.
- **Wave Polarization**: Electromagnetic waves can be polarized, meaning that the electric field oscillates in a specific plane as the wave propagates. Polarization can be linear, circular, or elliptical, depending on the orientation of the electric field vector.
- **Superposition**: Like other waves, classical electromagnetic waves obey the principle of superposition, which means that multiple waves can combine to form a new wave through constructive or destructive interference [4].
- **Energy Transport**: Electromagnetic waves carry energy as they propagate through space. The intensity of the wave, which represents the energy per unit area per unit time, is related to the amplitude of the electric and magnetic fields.
- **Wave Reflection, Refraction, and Diffraction**: Classical electromagnetic waves exhibit behaviors such as reflection when they encounter boundaries, refraction when passing from one medium to another, and diffraction when encountering obstacles. These phenomena can be described by classical wave optics.

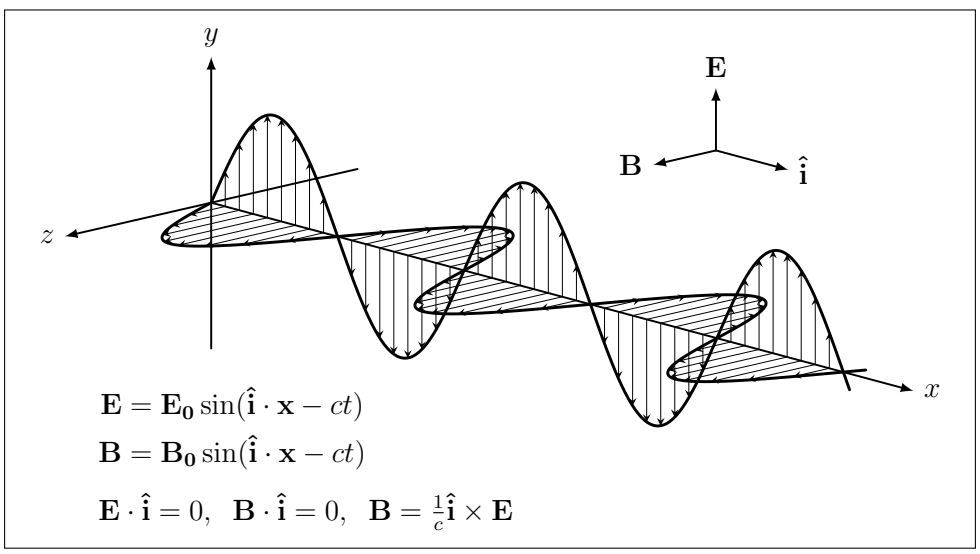

Figure 1: Propagation of an electromagnetic wave in the $\hat{\mathbf{i}}$ direction

It's important to note that while classical electromagnetic wave theory provides an accurate description of many electromagnetic phenomena, it does not account for the quantum nature of light, which is described by quantum electrodynamics (QED) [19]. In certain situations, particularly at very small scales or in the presence of intense electromagnetic fields, the quantum nature of light becomes significant, and classical wave theory may not fully explain the observed behavior.

6 Oliver Heaviside's Contributions to Maxwell's Equations

Oliver Heaviside made significant contributions to the understanding and reformulation of Maxwell's equations. His work helped simplify and clarify these equations, making them more accessible and useful for engineers and physicists. A few of Heaviside's important contributions to Maxwell's equations include [10, 15]:

- **Vector Notation**: Heaviside introduced the modern vector notation for Maxwell's equations, which greatly simplified their mathematical expression. This notation represented electric and magnetic fields as vectors, making it easier to write and manipulate the equations. Heaviside's notation is still widely used today [3].
- **Telegrapher's Equations**: Heaviside derived what are now known as the Telegrapher's Equations, which describe the behavior of electrical signals in transmission lines, such as telegraph cables. These equations are a simplified form of Maxwell's equations and are crucial for understanding the propagation of electromagnetic waves in transmission lines [20].
- **Concept of Impedance**: Heaviside introduced the concept of electrical impedance, which is a measure of how much a medium resists the flow of alternating current. He showed how impedance relates to the propagation of electrical signals in transmission lines and how it affects the reflection and transmission of signals [9].
- **Elimination of Quaternions**: Heaviside eliminated the use of quaternions, a complex mathematical system that Maxwell had initially used to describe electromagnetic phenomena. He replaced quaternions with the more intuitive vector notation, making Maxwell's equations more accessible to a wider audience [6].
- **Improved Understanding of Displacement Current**: Heaviside clarified the concept of displacement current, which is a term introduced by Maxwell to account for changing electric fields. Heaviside's work helped physicists better understand this concept and its role in the equations [14].
- **Operational Calculus**: Heaviside is also known for his significant contributions to the field of electrical engineering and the development of operational calculus. Operational calculus is a mathematical technique used to solve differential equations, especially those that arise in the study of electrical circuits and systems [17]. The operational calculus is also known as the Heaviside calculus in honor of his pioneering work in this area.

  Operational calculus is primarily concerned with transforming differential equations into algebraic equations, which are often easier to manipulate and solve. Heaviside's work helped simplify the analysis of complex linear systems, making it more accessible to engineers and scientists.

  Some of the key concepts and techniques associated with Heaviside's operational calculus include:

  - **Laplace Transforms**: Heaviside introduced the use of Laplace transforms to solve linear differential equations with initial conditions. This transformation converts a differential equation into an algebraic equation, simplifying the solution process [5].
  - **Differential Operators**: Heaviside introduced the notion of differential operators, such as the D-operator (d/dt), which allowed him to manipulate differential equations more effectively [13].

– **Heaviside Step Function**: Heaviside introduced the unit step function (also known as the Heaviside step function) as a way to represent and model sudden changes in electrical circuits. The Heaviside step function is shown in Definition 6.1 and Figures 2, 3 and 4.

Definition 6.1. Heaviside Step Function:

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

The Heaviside step function is shown in Figure 2.

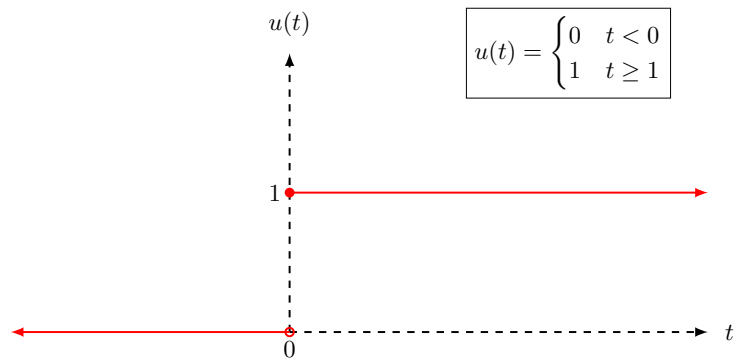

Figure 2: The Heaviside Function $u(t)$

More frequently we are interested in $u(t-a)$, which is sometimes written as $u_a(t)$ and is shown in Figure 3.

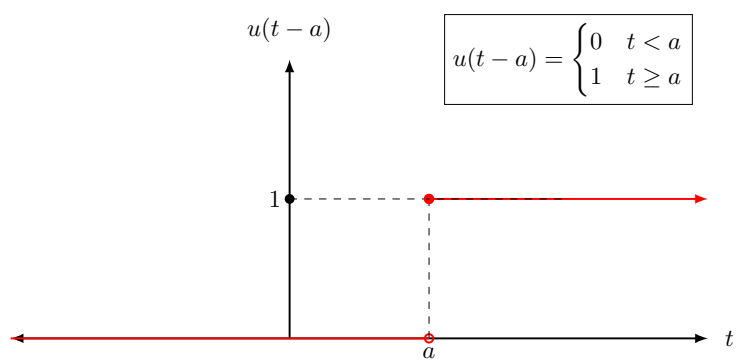

Figure 3: $u(t-a)$

Another useful version of the Heaviside step function is shown in Figure 4.

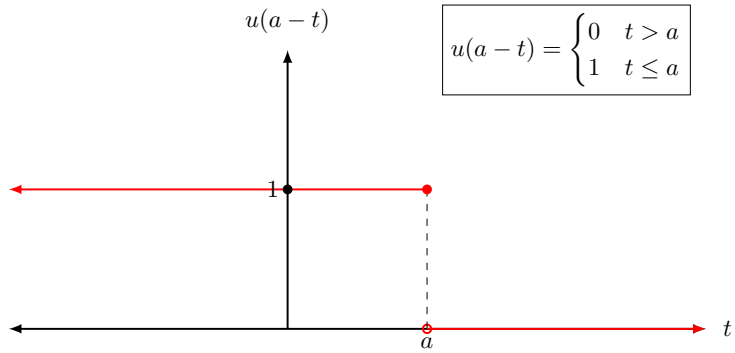

Figure 4: $u(a-t)$

> Heaviside's work in operational calculus and electrical engineering significantly influenced the development of modern control theory and signal processing. His contributions made it possible to analyze and design electrical and electronic systems more efficiently.
>
> Operational calculus remains an important tool in engineering and physics, especially in the analysis of linear time-invariant systems. While the formalism of operational calculus has evolved since Heaviside's time, his pioneering work laid the foundation for these developments and continues to be influential in various engineering disciplines.

Oliver Heaviside's contributions to Maxwell's equations were instrumental in simplifying and modernizing the mathematical description of electromagnetism. His work made it easier for engineers and scientists to apply these equations to practical problems and laid the foundation for many developments in electrical engineering and telecommunications.

7 Conclusions

8 Acknowledgements

Thanks to Larry Lang for suggesting that I add some text about Oliver Heaviside's important contributions to Maxwell's equations (see Section 6). Thanks also to rrogers (@rrogers@mathstodon.xyz) for pointing out Heaviside's contributions to the development of operational calculus.

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